

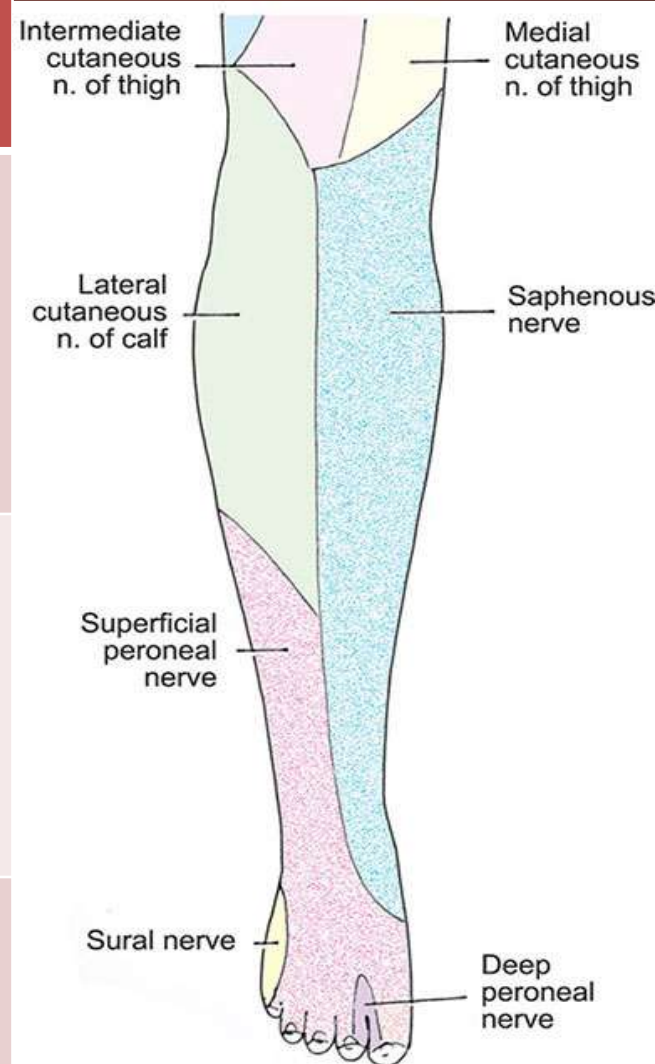
Anterior Compartment of the Leg (Extensors)

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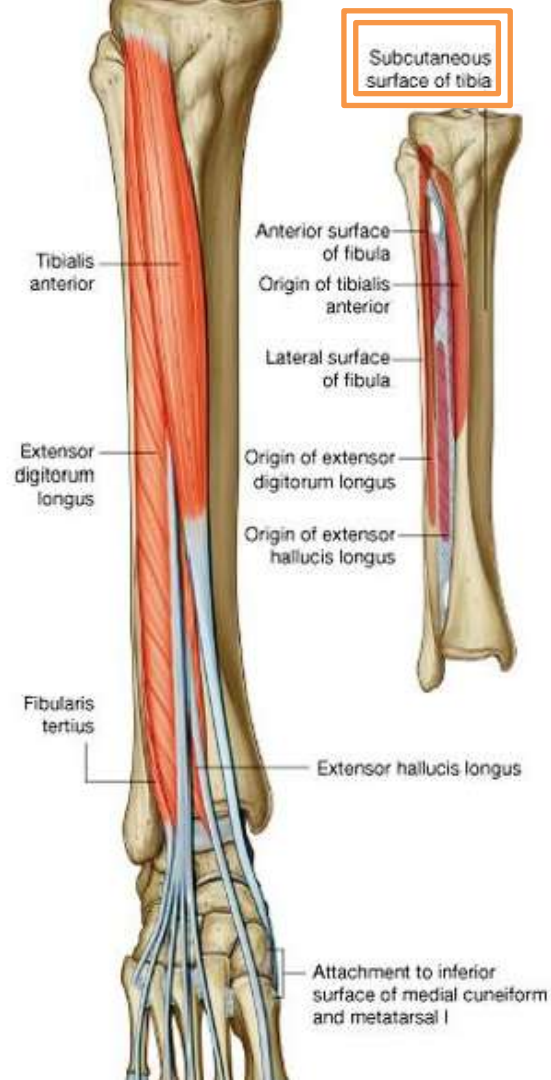


Cutaneous nerve	Origin	Skin supplied
1. Saphenous nerve	branch of femoral nerve	To skin of anteromedial surface & posteromedial surface of the leg. (it ends on medial side of the foot).
2. Lateral cutaneous nerve of the calf	Branch of common peroneal nerve .	To skin on the upper part of the antero-lateral surface of the leg. & the upper part of postero-lateral surface of the leg.
3. Superficial peroneal nerve	Terminal branch of common peroneal nerve .	To skin of lower part of anterolateral surface of the leg.





Foot extension = dorsi flexion
Foot flexion = plantar flexion



Observe the medial subcutaneous surface of tibia is free of muscles

Content:

Foot & toes extensors

❖ **Muscles:**

1-Tibialis anterior 2- Extensor digitorum longus
3- peroneus tertius 4-Extensor hallucis longus.

❖ **Blood supply:**

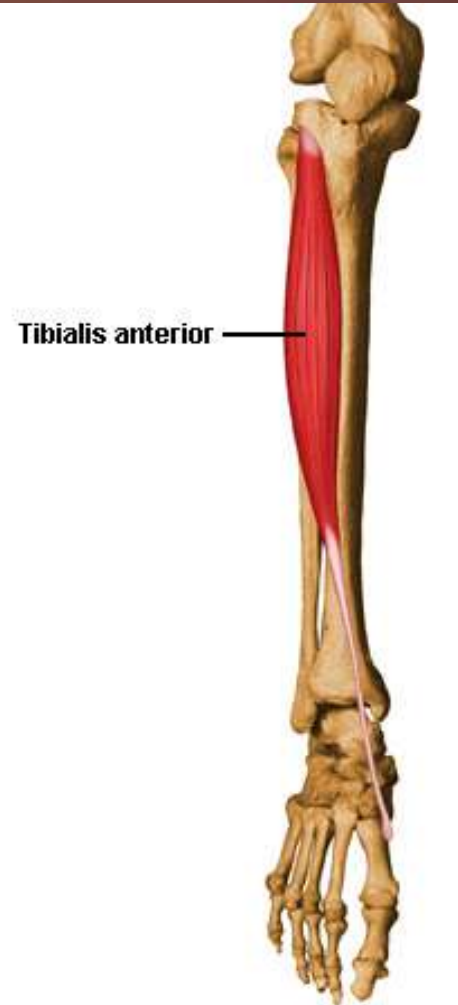
Anterior tibial artery

❖ **Nerve supply:**

Deep peroneal nerve

1-Tibialis anterior

- O** : From the upper $\frac{1}{2}$ of the **lateral surface of the tibia** & from the interosseous membrane.
- I** : Tendon passes deep to both extensor retinacula & is attached to **medial cuneiform bone & base of the 1st metatarsal bone**.
- N** : **Deep Peroneal nerve**.
- A** : Extends(**dorsiflexion**) foot at ankle joint + Inverts the foot + assist in holding up medial longitudinal arch of foot.



2-Extensor hallucis longus

It lies next to and overlapped by tibialis anterior muscle.

O: from **middle** $\frac{1}{2}$ of **anterior surface of fibula** & interosseous membrane.

I :its tendon appears between the tibialis anterior tendon & EDL tendon in the lower half of leg,,
into base of distal phalanx of big toe.

N: **deep peroneal nerve**

A: 1) Extends the big toe
2) Extends the foot at ankle joint
3) Assists in **inversion** of the foot at subtalar & transverse tarsal.



3-Extensor digitorum longus

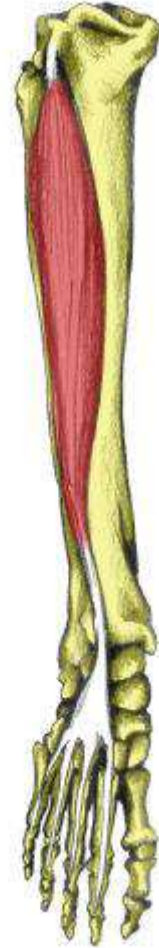
Is the most posterior & lateral muscle in the anterior group

O : from the **upper 2/3** of the **anterior surface of fibula & interosseous membrane**.

I : its tendon passes behind extensor retinacula, divides into 4 tendon , **inserted into** dorsal surfaces(extensor expansion) of the bases of **middle & distal** phalanges of the lateral 4 toes.

N : **deep peroneal nerve**

A : 1) **Extends(dorsiflexes)** foot at ankle joint.
2) **Extends** the lateral 4 toes.



4-Peroneus Tertius

❑ This muscle is part of extensor digitorum longus.

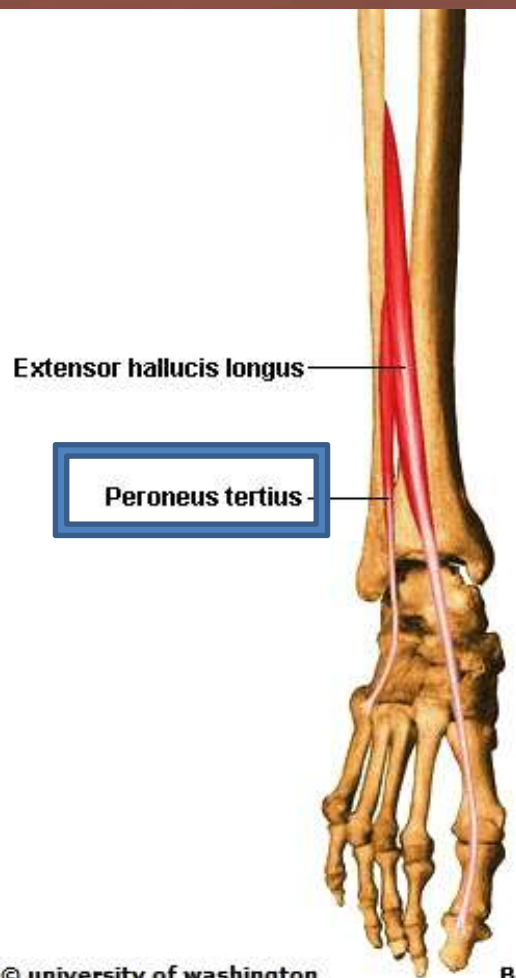
O : from the lower 1/3 of anterior surface of fibula & interosseous membrane.

I : its tendon follows the tendons of extensor digitorum longus behind extensor retinaculum (shared synovial sheath), inserted into **medial side of dorsal aspect of the base of the 5th metatarsal bone**. (like p.brevis)

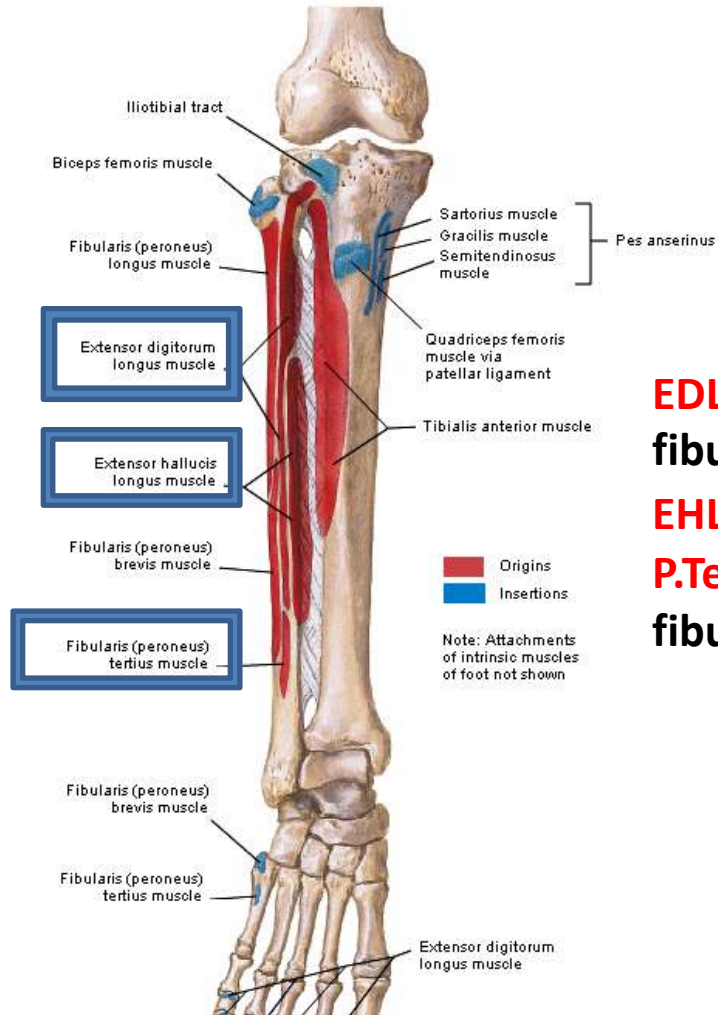
N : **deep peroneal nerve**

A : **Extends the foot at the ankle joint + Everts the foot at subtalar & transverse tarsal joints.**





Attachments of Muscles of Leg: Anterior View



EDL: upper $\frac{2}{3}$ of the anterior surface of fibula.

EHL: middle $\frac{1}{2}$ of anterior surface of fibula

P.Tertius: lower $\frac{1}{3}$ of anterior surface of fibula

**Extensor
Digitorum Longus**

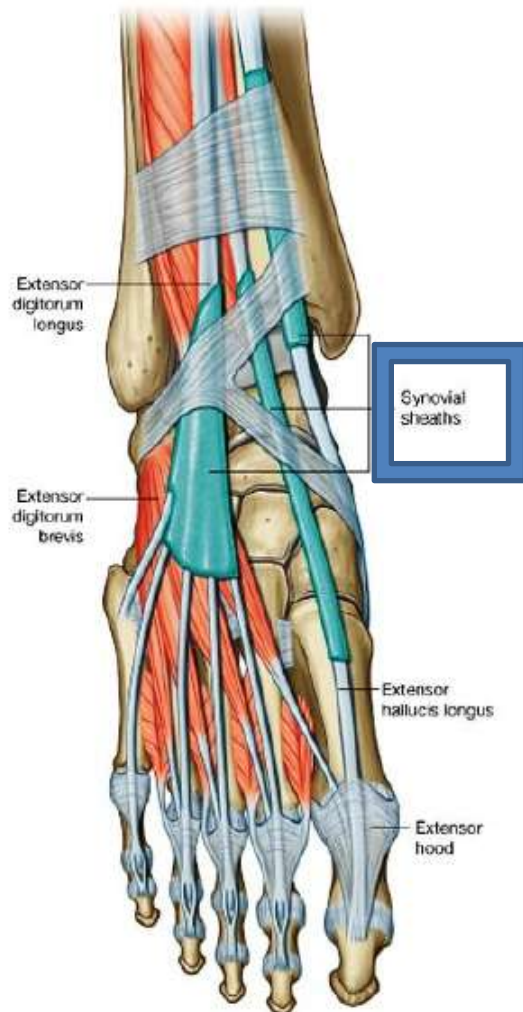
**Tibialis
Anterior**

**Extensor
Hallucis Longus**



teachmeanatomy

The #1 Applied Human Anatomy Site on the Web





- ❖ Both tibialis anterior and tibialis posterior producing **inversion** of foot. +assistance (EHL).
- ❖ Peronius longus, previus, tertius producing **Eversion** of foot.

❑ Peroneus longus & previus → plantar .
Flexors
(flexors)

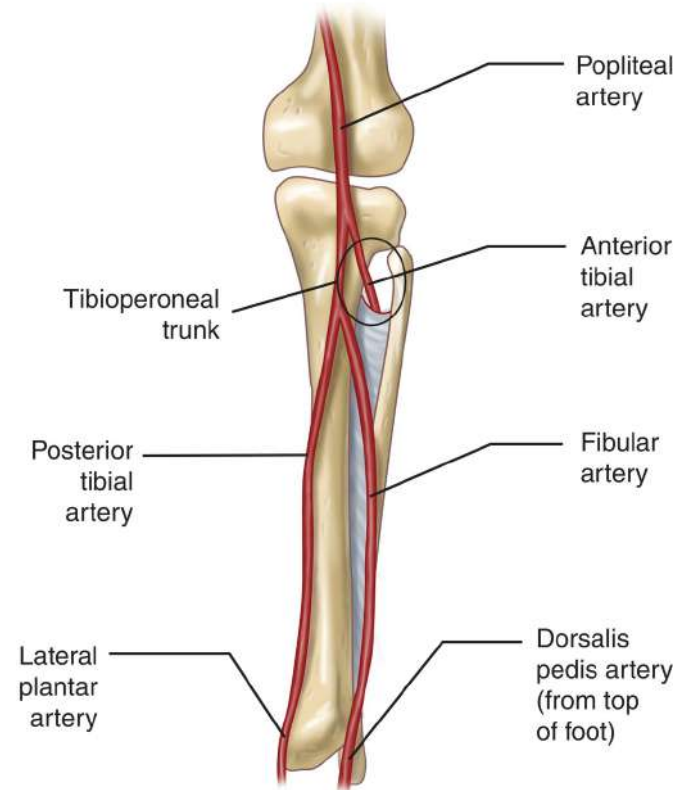
❑ Peroneus tertius → dorsi flexors
(extension)

❑ All peroneal muscles are Evertors for the foot.

Artery of anterior compartment of the leg:

❖ Anterior tibial artery

- ❖ **Beginning:** Smaller of terminal branches of popliteal artery at the level of lower border of popliteus.
- ❖ **Course:** It passes to anterior compartment of the leg through **opening in interosseus membrane**.
 - Then it descends on interosseous membrane together with **deep peroneal nerve**.
- ❖ **End:** in front of the ankle joint the artery becomes **dorsalis bedis artery**. مهم



- **Branches:**

1- **muscular branches:** to neighboring muscles.

2- **anastomotic branches :** with branches of other arteries around the knee & ankle joints.

Feeling pulsation of dorsalis pedis artery



It lies between the **tendons of extensor hallucis longus & extensor digitorum longus** , midway between the medial and lateral malleoli on the front of the ankle .It can easily be felt by living subject.



Using ultrasound to measure blood flow in dorsalis pedis artery.



Applied

❑ Popliteal pulse:

- Difficult to feel because popliteal artery is deeply placed in fossa.
- Best performed with person in prone position with knee flexed to relax popliteal fascia and hamstrings.
- Pulsation **best felt in inferior part of the fossa where pop.artery is related to tibia.**



Nerve of anterior compartment of the leg

□ Deep peroneal nerve

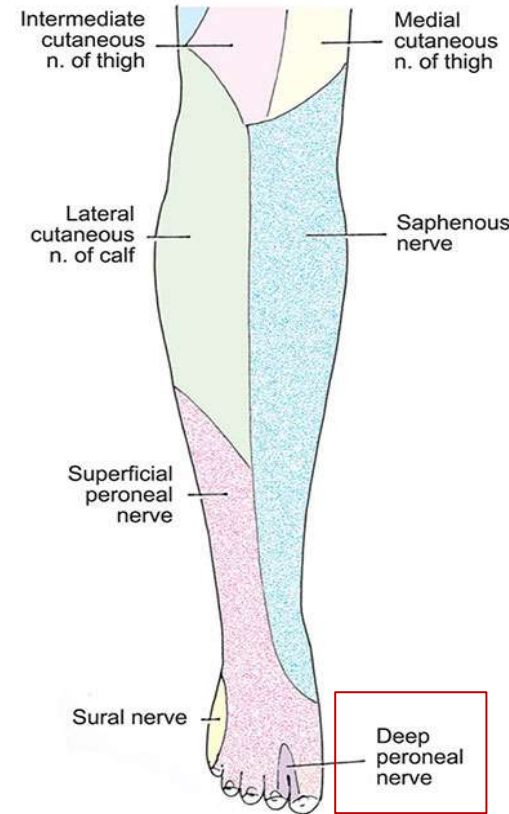
- **Origin:** It begins in the **substance of peroneus longus muscle** as one of the two terminal branches of common peroneal nerve.
- **Course:**
 - The nerve enters the anterior compartment of leg.
 - It accompanies the **anterior tibial artery**.
 - Both pass behind extensor retinaculum to enter dorsum of foot. **The nerve is lateral to the dorsalis pedis artery.**
- **End:** it divides into terminal **medial & lateral branches**.

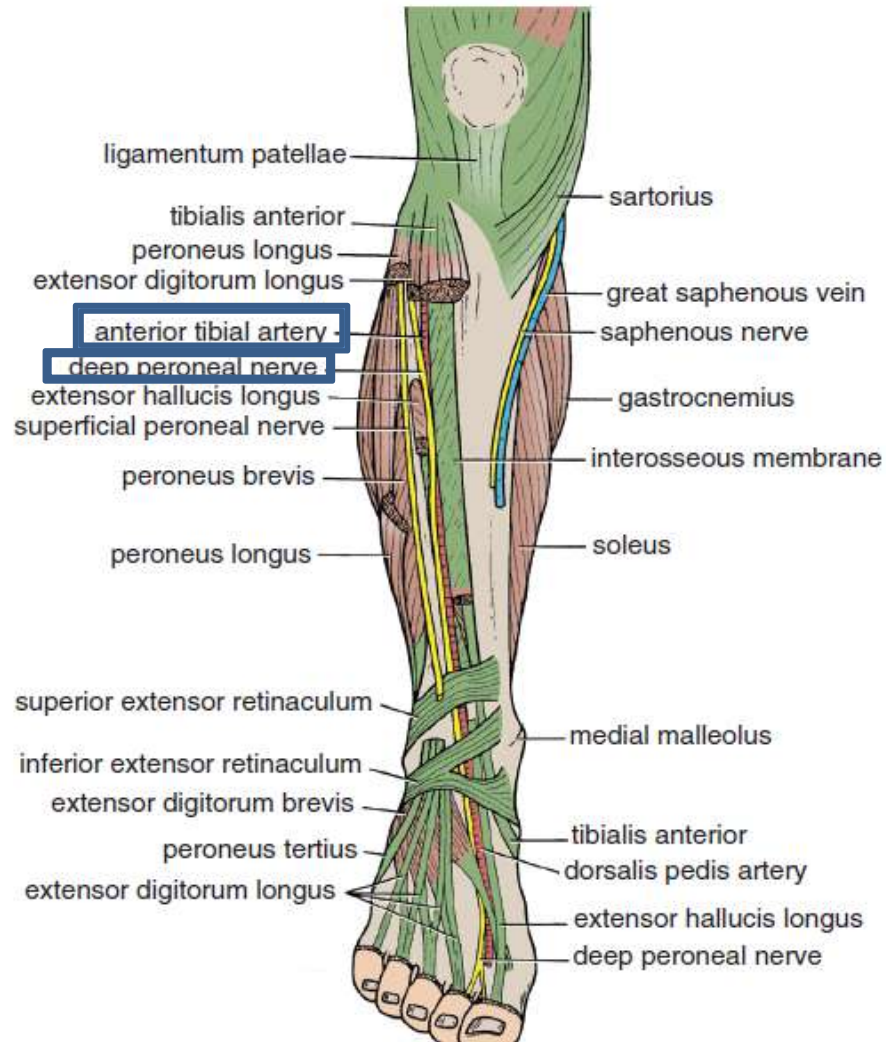
❑ Branches:

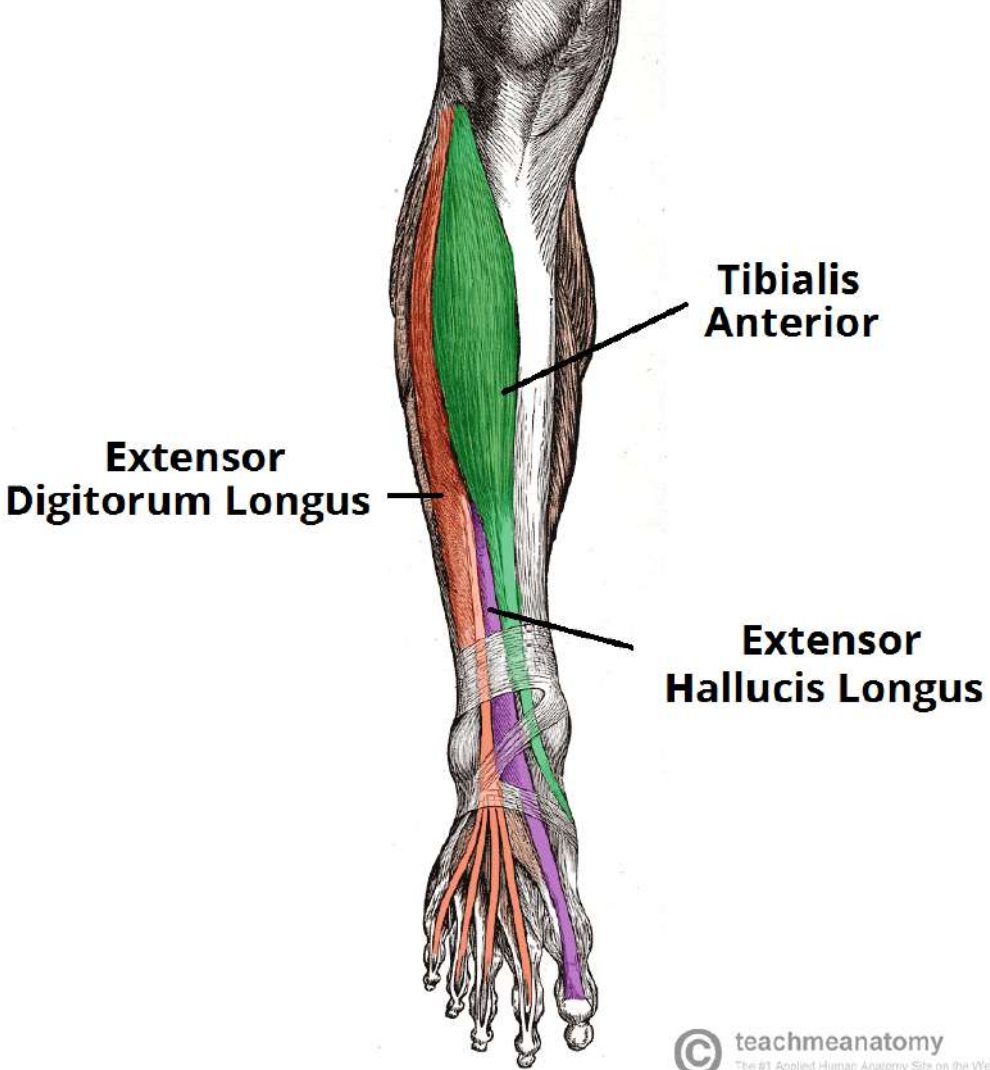
1-Cutaneous: skin of adjacent sides of the big & 2nd toes (medial plantar n)

2-Muscular: to all muscles on the anterior compartment of leg
+ **Extensor digitorum brevis** on the dorsum of foot.

3-Articular: to ankle joint.







Region of the ankle:

Clinically, ankle is a common site for fractures, sprains and dislocations.

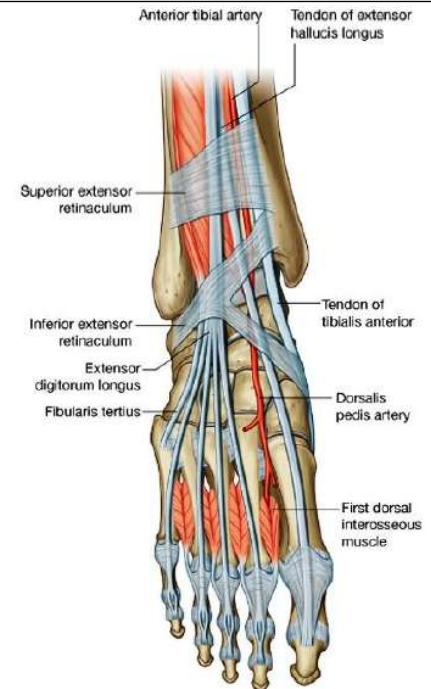


Ankle region :Anterior aspect.

- **Extensor retinaculum**

1) **Superior extensor retinaculum**
retinaculum: above ankle joint, attached to ant. borders of tibia & fibula.

2) **Inferior extensor retinaculum:** looks like **letter of (Y)**, more important, lying close to ankle joint.



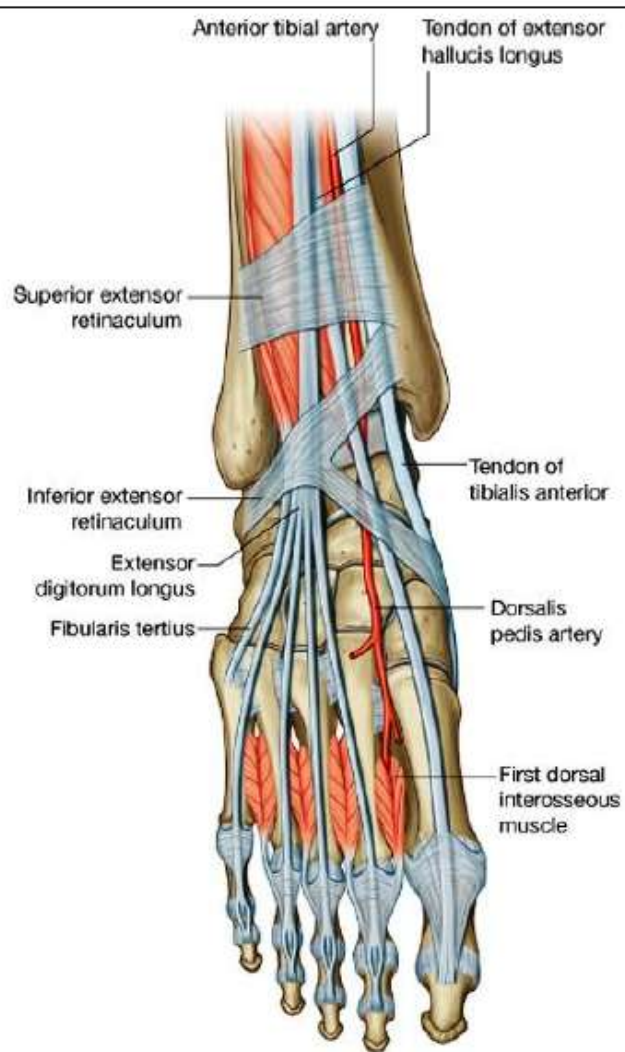
Structures that pass beneath(under) or through the extensor retinacula of the foot.

From medial to lateral

- 1-Tibialis anterior
- 2-Extensor hallucis longus tendon.
- 3-Anterior tibial artery with vnae comitants
- 4-Deep peroneal nerve
- 5-Extensor digitorum longus tendon
- 6- Peroneus tertius.

Tom has a nice dog & pig

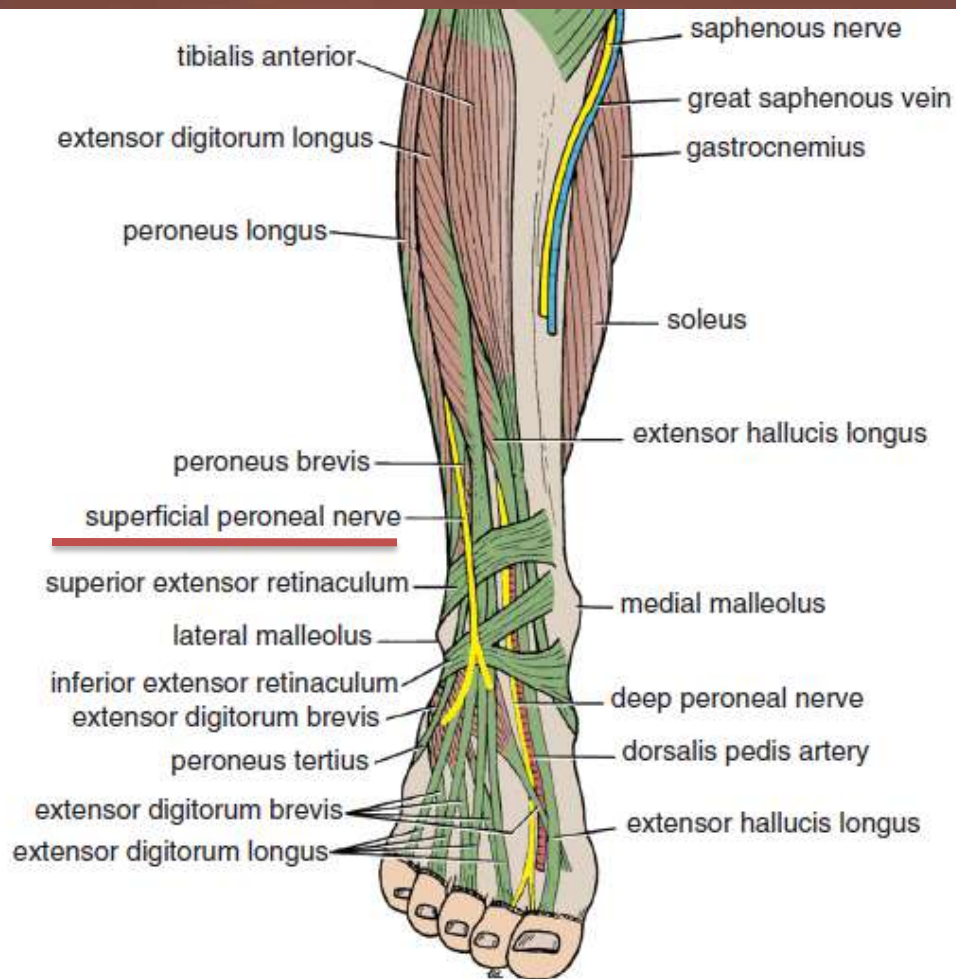
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Structures passing
behind extensor
retinaculum

Structures that pass anterior (superficial) to extensor retinaculum of the foot.

- 1-Saphenous nerve and great saphenous vein (infront of medial malleolus)**
- 2-Superficial peroneal nerve**



Observe superficial peroneal nerve passing superficial to extensor retinaculum.

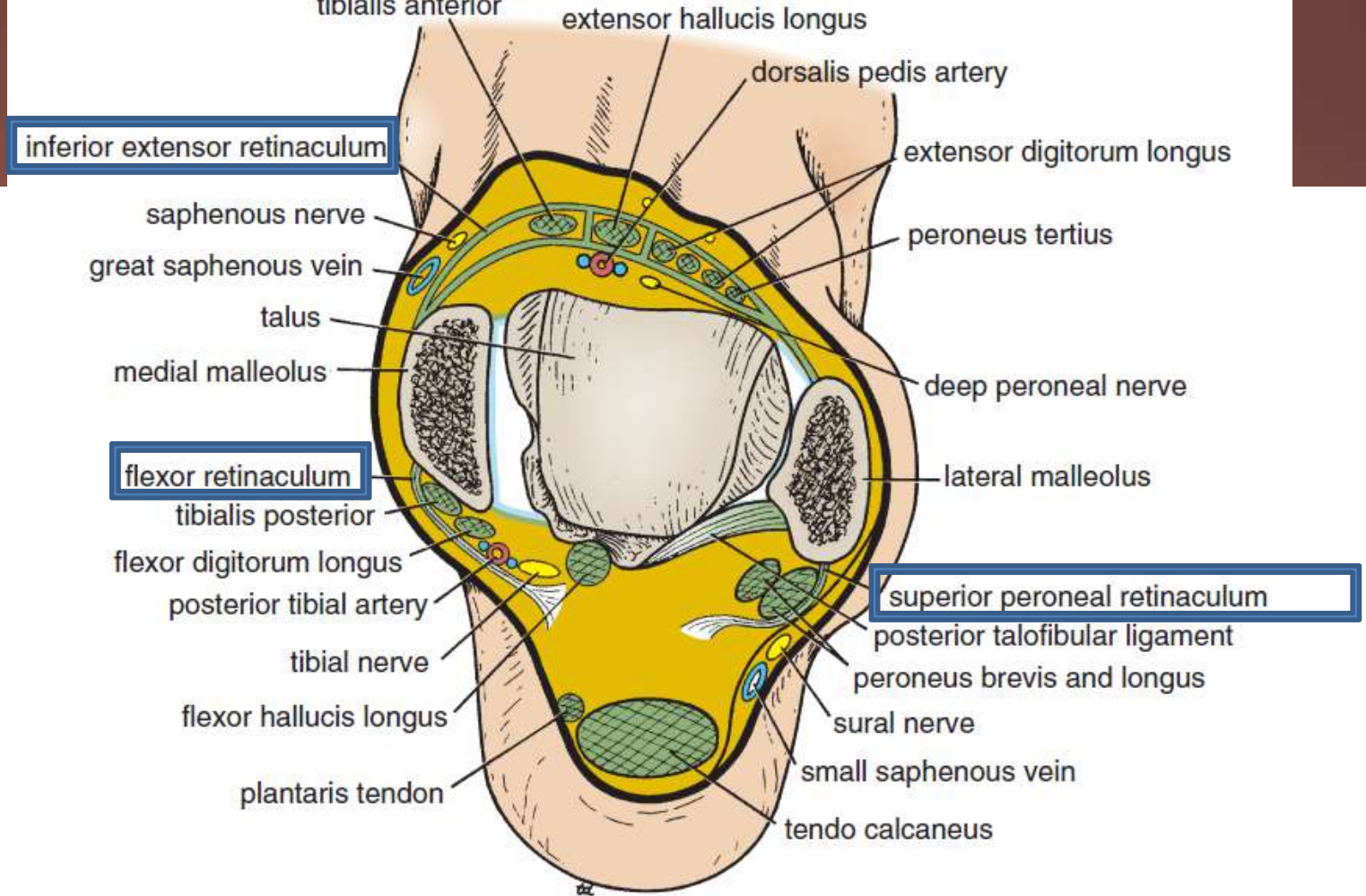


FIGURE 10.48 Relations of the right ankle joint.



Formative

What is the nerve supply of the anterior leg?

- a. Superficial peroneal nerve
- b. Deep peroneal nerve
- c. Obturator nerve
- d. Tibial nerve

What is the main artery supplying the anterior leg?

- a. Anterior tibial
- b. Posterior tibial
- c. Peroneal
- d. obturator

What is the name of cutaneous nerve supplying antromedial aspect of the leg?

- a. Saphenous nerve
- b. Superficial peroneal nerve
- c. Medial cutaneous nerve of calf
- d. Sural

Which of the following structures passes deep to extensor retinaculum of the ankle?

- a. Tibialis posterior
- b. Tibialis anterior
- a. Flexor hallucis longus
- b. Tibial nerve

- Which of the following structures does not pass deep to Flexor retinaculum of the ankle?
 - a. Tibial nerve.
 - b. Tibialis posterior tendon
 - c. Tibialis anterior tendon
 - d. Posterior tibial artery

Which of the following muscles is evertor of the foot?

- a. Tibialis anterior
- b. Tibialis posterior
- c. Peroneus longus
- d. Extensor hallucis longus



الحياة مستمرة

سواء ضحكت أم بكيت
فلا تُحْمَلْ نفسك هموماً
لن تستفيد منها وابتسم